

UNIT 7

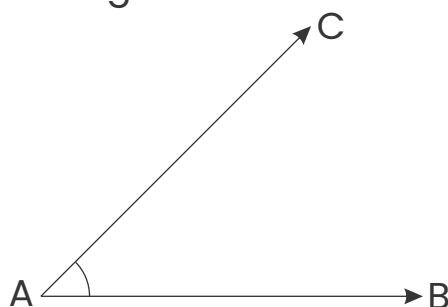


GEOMETRY



ANGLES

Recall an angle and recognize acute, right, obtuse, straight and reflex angle. An angle is formed when two straight lines or rays meet at a common endpoint. In the given figure, point A is called **vertex**. AB or AC are called its **arms** or **sides**. AB is **initial arm** and AC are **terminal arm**. It is named as angle BAC or angle CAB. " $\angle$ " is the symbol for an angle. It is written as $\angle$ BAC or $\angle$ CAB. It is read as angle BAC or angle CAB.



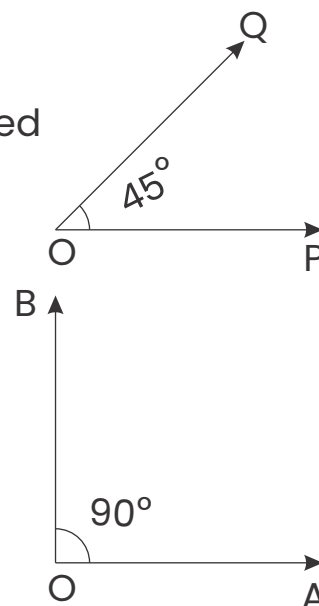
Kinds of an Angle

(i) Acute angle:

An angle whose measure is less than 90° is called an acute angle.

Here $m \angle POQ = 45^\circ$ (m stands for measure)

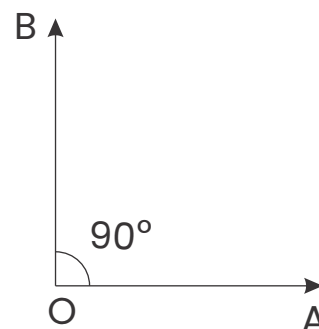
Therefore $\angle POQ$ is an acute angle



(ii) Right angle:

An angle whose measure is 90° is called right angle.

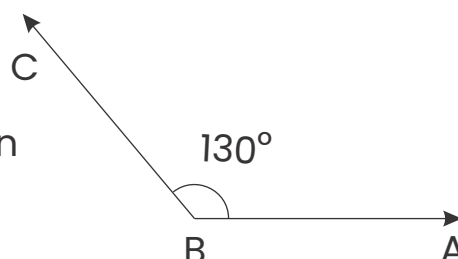
Here $m \angle AOB = 90^\circ$ Therefore $\angle AOB$ is a right angle.



(iii) Obtuse angle:

An angle whose measure is greater than 90° but less than 180° is called an obtuse angle.

Here $m \angle ABC = 130^\circ$ Therefore $\angle ABC$ is an obtuse angle.

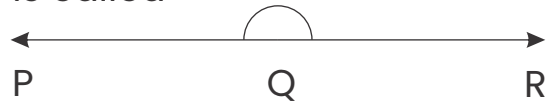


(iv) Straight angle:

An angle whose measure is 180° is called a straight angle.

Here $m \angle PQR = 180^\circ$

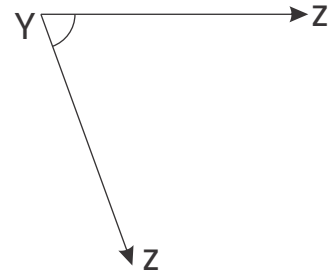
Therefore $\angle ABC$ is a straight angle

**(v) Reflex angle:**

An angle whose measure is greater than 180° but less than 360° is called 290° a reflex angle.

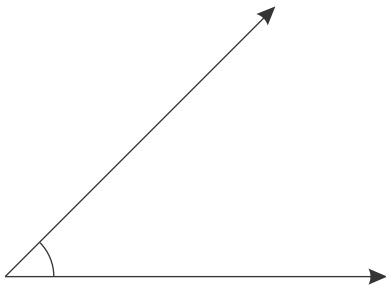
In the given figure $m \angle XYZ = a = 290^\circ$

Therefore, XYZ is a reflex angle.

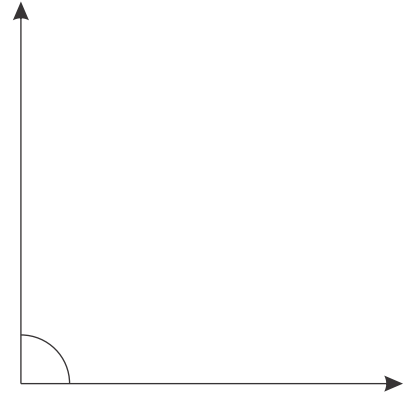
**EXERCISE 7.1**

Q1. Measure the angles and identify the type of each angle.

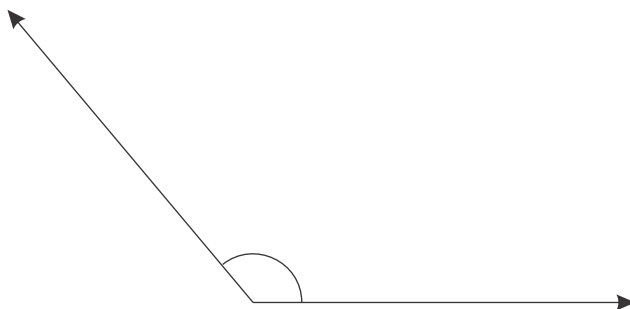
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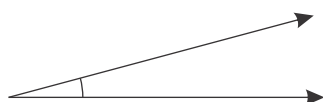


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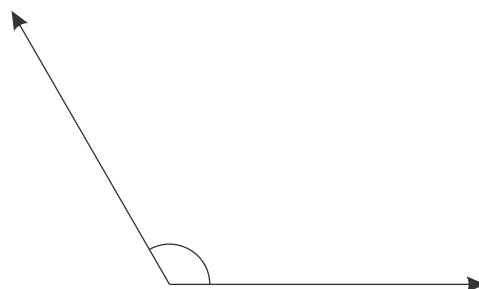


Q2. Identify the angles as acute, right, obtuse, straight or reflex angle.

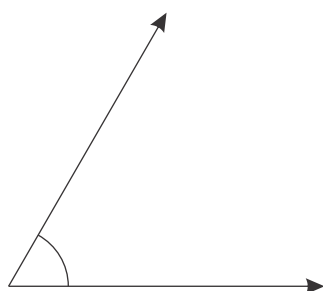
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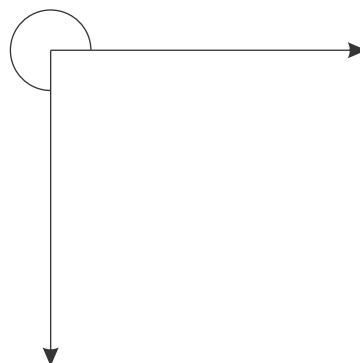
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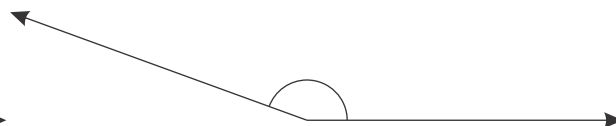
(iv)



(v)



(vi)



Use protractor to construct a right angle, a straight angle and reflex angles of different measures.

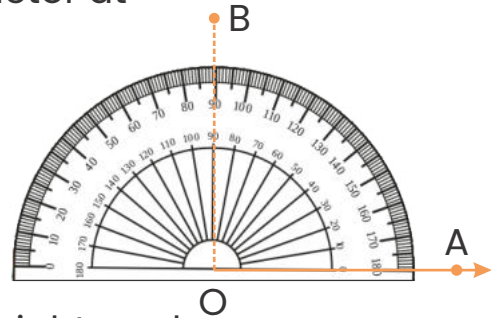
1. To construct a right angle with the help of protractor.

Steps of construction:

- (i) Draw an $\overrightarrow{OA}$
- (ii) Place the centre point of the protractor at point O along $\overrightarrow{OA}$
- (iii) Read the measurement at the protractor from the side where its zero mark lies on $\overrightarrow{OA}$. Mark the point B at 90° .

- (iv) Draw $\overrightarrow{OB}$ shown in the picture.

Thus $m\angle AOB = 90^\circ$ and it is the required right angle.

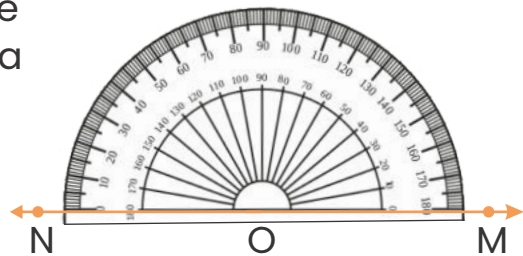


2. To construct a straight angle with the help of protractor.

Steps of construction:

- (i) Draw a ray $\overrightarrow{OM}$.
- (ii) Place the centre point of the protractor on point O along $\overrightarrow{OM}$.
- (iii) Read the protractor from the side where its zero mark lies on $\overrightarrow{OM}$. Mark a point N at 180° .

- (iv) Now draw $\overrightarrow{ON}$ as shown in the picture. Thus $m\angle MON = 180^\circ$ which is the required straight angle.



3. Using the protractor to construct reflex angle of different measures.

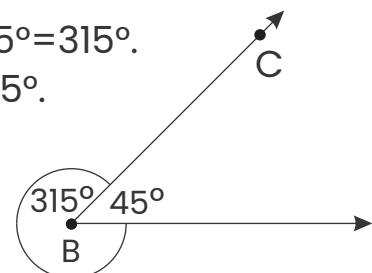
Steps of construction:

Let us draw and measure a reflex angle of 315° using a protractor.

We know that one complete turn = 360° .

Therefore the measure of reflex $ABC = 360^\circ - 45^\circ = 315^\circ$.

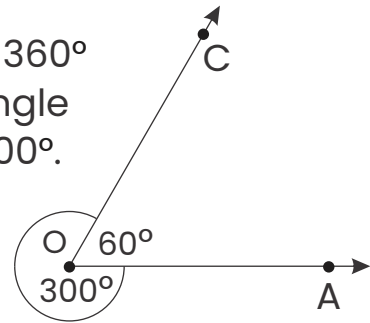
We first draw and measure the acute $ABC = 45^\circ$.



Example: Construct a reflex angle of 300° .

Steps of construction:

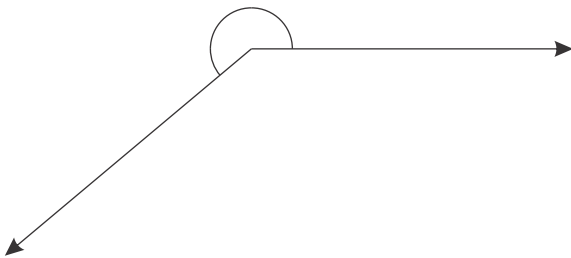
We subtract the given measure of 300° from 360° i.e. $360^\circ - 300^\circ = 60$. Now we draw an acute angle of 60° as to get the required reflex angle of 300° . AOC is the required reflex angle. In this way, we can draw a reflex angle of different measure.



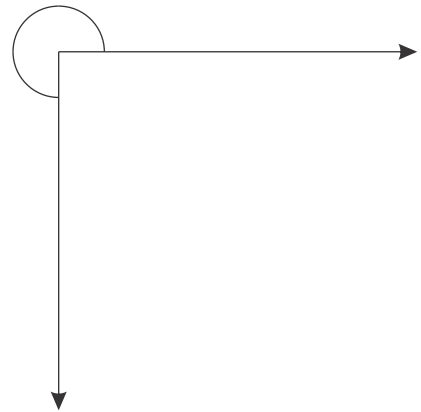
EXERCISE 7.2

Q1. Measure the reflex angles by using protractor.

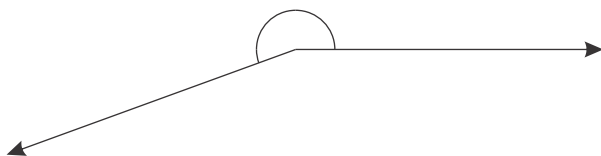
(i)



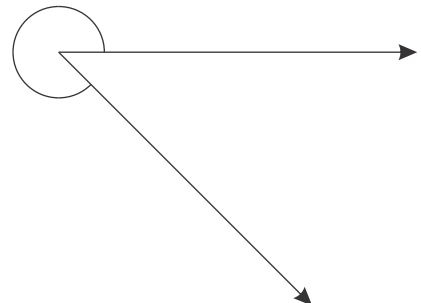
(ii)



(iii)



(iv)



Q2. Draw and label the following angles:

1. PQR (straight angle)

Solution:

2. MNO (right angle)

Solution:

3. XYZ (reflex angle)

Solution:

4. PQR (obtuse angle)

Solution:

5. ABC (acute angle)

Solution:

Q3. Using protractor, draw the following:

(i) Reflex PQR of 280°

Solution:

(ii) Reflex ABC of 320°

Solution:

(iii) Reflex GHI of 350°

Solution:

Adjacent, Complementary and Supplementary angles

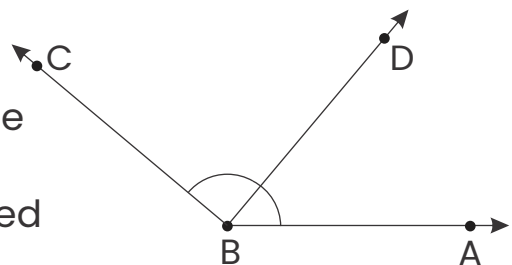
(i) Adjacent Angles:

Two angles are said to be adjacent angles if they have a common vertex, and a common arm.

Look at the figure. We have two angles: (i) ABD and (ii) DBC.

There is a common vertex at point B and a common arm $\overrightarrow{BD}$. The other two arms $\overrightarrow{BA}$ and $\overrightarrow{BC}$ of the angles are on the opposite sides of the common arm $\overrightarrow{BD}$.

Two such angles: ABD and DBC are called adjacent angles.



(ii) Complementary Angles:

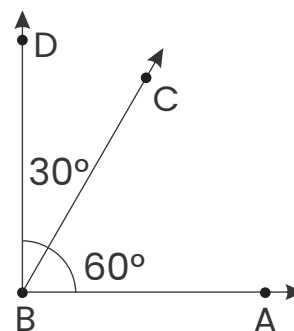
Two angles whose measures have a sum of 90° are called complementary angles. In the figure, 60° and 30° are two complementary angles because $m\angle ABC + m\angle CBD = 60^\circ + 30^\circ = 90^\circ$.

Each angle is called a complement of the other.

So,

ABC is complement of CBD and = CBD

is complement of ABC



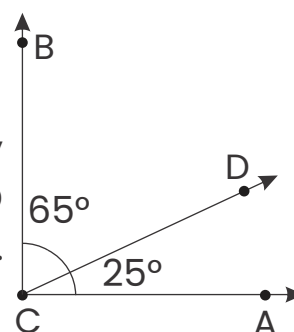
Example:

Find the complement of 65° ?

Solution:

m BCD is 65° , then measure of its complementary angle m $\angle ACD$ will be $90^\circ - 65^\circ = 25^\circ$ Because BCD and ACD and $\angle ACD$ are two complementary angles.

$$m\angle BCD + m\angle ACD = 65^\circ + 25^\circ = 90^\circ$$



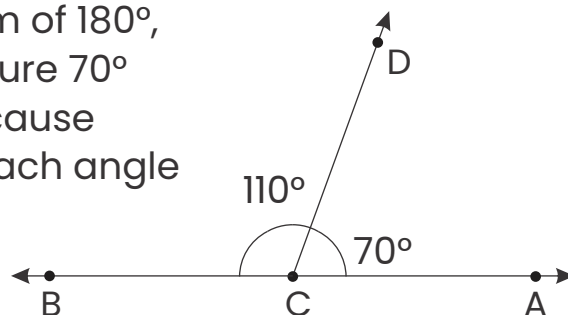
(iii) Supplementary Angles

Two angles whose measure have a sum of 180° , are called supplementary angles. In figure 70° and 110° are supplementary angles because

$m\angle ACD + m\angle DCB = 70^\circ + 110^\circ = 180^\circ$ Each angle is called supplement of the other.

Then ACD is supplement of DCB or

DCB is supplement of ACD.



Example:

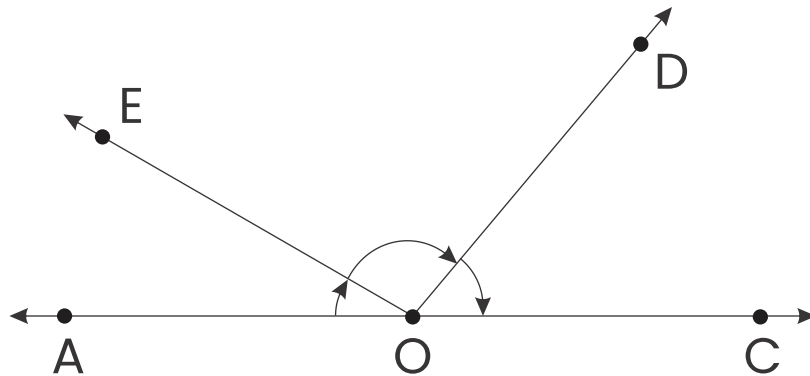
Find supplement of the given angle of 140° .

Solution:

$$\text{Supplement of } 140^\circ = 180^\circ - 140^\circ = 40^\circ$$

EXERCISE 7.3

Q1. Look at the figure and answer the following:



Is AOE adjacent to DOE? Why or why not?

Solution:

Is AOD adjacent to COD?

Solution:

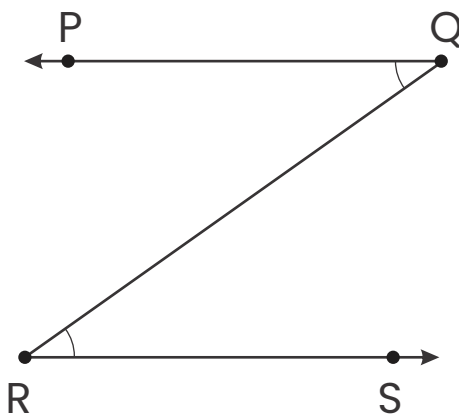
Is AOE adjacent to AOD?

Solution:

Is DOE adjacent to EOC?

Solution:

Q2. Is PQR adjacent to QRS? Why or why not?



Solution:

Q3. Find the complement of each of the following angles:

35°

Solution:

58°

Solution:

82°

Solution:

Q4. Find the supplement of each of the following angles:

78°

Solution:

135°

Solution:

63°

Solution:

Q5. Identify which of the following pair of angles are complementary and which are supplementary.

42°, 48°

Solution:

37°, 143°

Solution:

138°, 42°

Solution:

Q6.(i) Find the angle which is equal to its complement.

Solution:

(ii) Find the angle which is equal to its supplement.

Solution:

Q7. Can two angles be supplementary, if both of them are:

a. Obtuse

Solution:

b. Acute

Solution:

c. Right

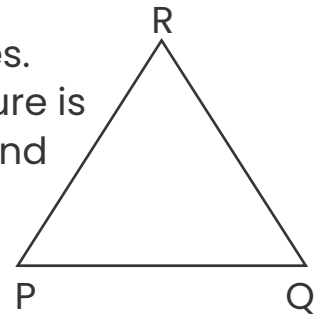
Solution:

ANGLES

Definition of a triangle:

A triangle is a plane closed figure with three sides. The symbol of triangle is $\triangle$. Given figure is $\triangle PQR$. Points P, Q, and R are three vertices. $\overline{PQ}$, $\overline{QR}$ and $\overline{RP}$ are three sides. Angle PQR, QRP and RPQ are three angles.

The sum of all three angles of a triangle is equal to 180° .

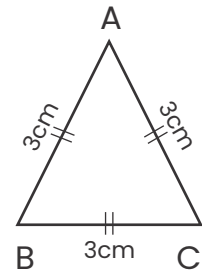
**Types Of Triangles With Respect To Their Sides**

According to the sides of a triangle, there are three types of a triangle.

(i) Equilateral Triangle

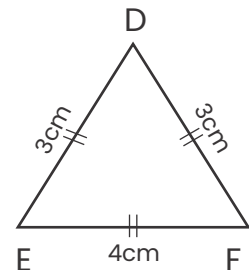
It is a triangle having three sides equal in length. All of its angles measure 60° .

ABC is an equilateral triangle. $\overline{mAB} = \overline{mBC} = \overline{mCA}$
 $m\angle A = m\angle B = m\angle C = 60^\circ$

**(ii) Isosceles Triangle**

It is a triangle having two sides equal. The angles opposite to the equal sides of an isosceles triangle are also equal.

DEF is an isosceles triangle. $\overline{mDE} = \overline{mDF}$
 $m\angle E = m\angle F$

**(iii) Scalene Triangle**

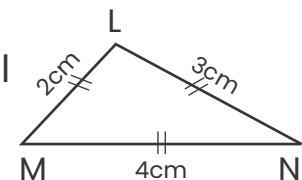
It is a triangle of different all sides in length. It's all angles are also different.

LMN

is a scalene triangle. Because

$\overline{mLM} \neq \overline{mMN} \neq \overline{mLN}$

$m\angle L \neq m\angle M \neq m\angle N$



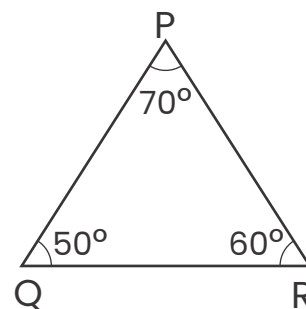
Types Of Triangles With Respect To Their Angles

According to the angles of a triangle, there are three types of a triangle.

(1) Acute-angled Triangle

If all three angles of a triangle are acute angles, then the triangle is called an acute-angled triangle.

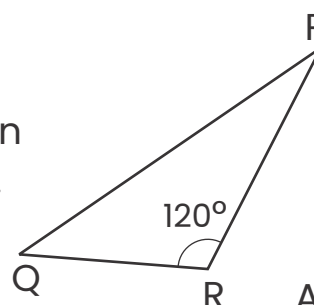
PQR is an acute-angled triangle, in which P, Q and R are acute angles.



(2) Obtuse-angled Triangle

If one angle of a triangle is an obtuse angle, then the triangle is called an obtuse-angled triangle.

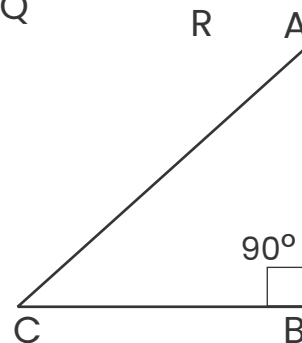
PQR is an obtuse-angled triangle, in which QRP is an obtuse angle.



(3) Right-angled Triangle

If one angle of a triangle is a right angle, then the triangle is called a right-angled triangle.

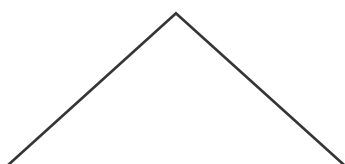
ABC is a right-angled triangle, in which $m\angle ABC = 90^\circ$



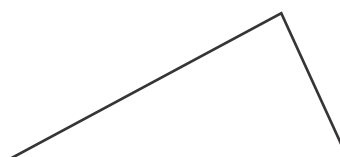
EXERCISE 7.4

Q1. Measure the sides of the following triangles and then write their names.

(i)



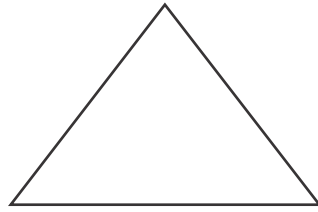
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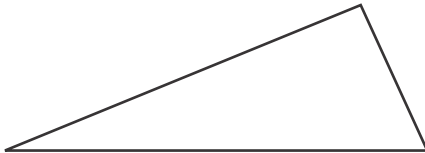


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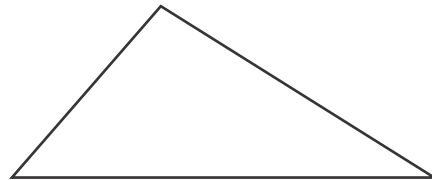


Q2. Use protractor, measure the angles of the following triangles and write their names.

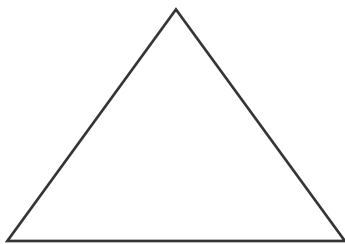
(i)



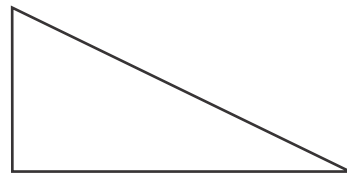
(ii)



(iii)



(iv)



Use compasses and straight edge to construct equilateral, isosceles and scalene triangles when three sides are given.

A. Using pair of compasses and ruler to construct an equilateral triangle when the measure of each side is given.

Example 1. Construct an equilateral triangle PQR such that its each side is 3.8 cm.

Given: Each side of triangle is 3.8 cm. Since it is an equilateral triangle hence measure of each angle will be 60° .

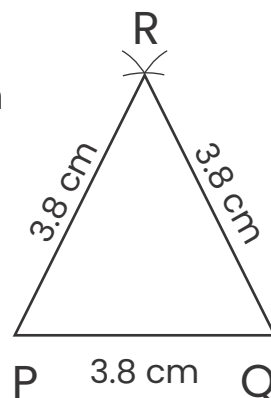
Steps of construction:

Step 1: Draw a line segment 3.8 cm length with the help of ruler. Name it as $\overline{PQ}$.

Step 2: Take P and Q as centres, open pair of compass to measure 3.8 cm. Draw an arc of radius 3.8 cm from point P and another from point Q, such as to intersect each other. Mark the point where two arcs meet as R.

Step 3: Draw $\overline{PR}$ and $\overline{QR}$.

Thus PQR is the required equilateral triangle.



B. Using pair of compasses and ruler to construct an isosceles triangle when measure of sides are given.

Example 2. Construct a triangle DEF when $mDE = mDF = 4.5$ cm and $mEF = 3.5$ cm

Steps of construction:

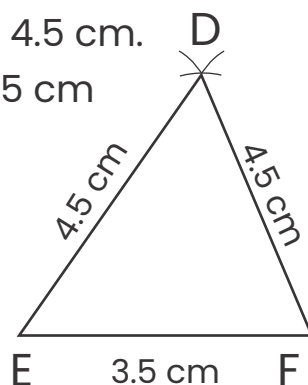
Step I: Use a ruler and draw EF of 3.5 cm length.

Step II: Open the pair of compasses to measure 4.5 cm. Take E and F as centre. Draw two arcs each of 4.5 cm for E and F, such as to intersect other.

Step III: Mark the point where two arcs intersect as D.

Step IV: Draw DE and DF

Thus, DEF is the required isosceles triangle.



- C.** To construct a scalene triangle, the measures of all three sides are given. We use the following steps which are given in the example.

Example 3. Construct a triangle ABC when $m AB = 5\text{cm}$ $m AC = 4\text{cm}$ and $m BC = 3\text{cm}$

Steps of construction:

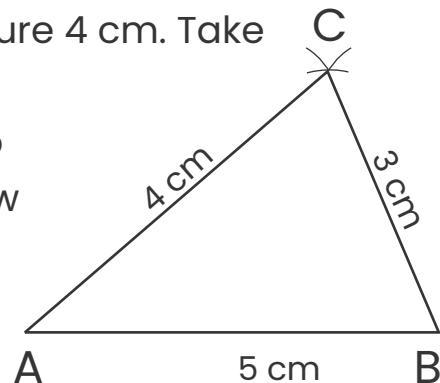
Step I: Use a ruler and draw AB of 5 cm in length.

Step II: Open pair of compasses to measure 4 cm. Take A as centre and draw an arc.

Step III: Again open pair of compasses to measure 3 cm. Take B as centre and draw another arc to meet the previous arc.

Step IV: Mark the point where two arcs meet; mark as C. Draw AC and BC.

Thus, ABC is the required scalene triangle.



Use a protractor and ruler to construct equilateral, isosceles and scalene triangles, when two angles and included side are given. Measure the length of the remaining two sides and one angle of the triangle.

We use the following steps given in the examples.

Example 1. Construct an equilateral triangle when $m \angle PQR = 60^\circ = m \angle QPR$ and $m PQ = 4\text{cm}$

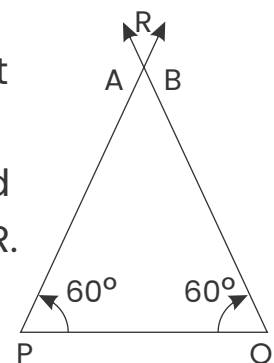
Steps of construction:

Step I: Draw a line segment PQ measuring 4 cm.

Step II: Place the central point of the protractor at point P. Read from Q and mark point A at 60°

Step III: Place the central point of the protractor at point Q. Read from P and mark point B at 60°

Step IV: From points P and Q draw the rays PA and QB, so that the rays intersect each other at point R. Now PQR is the required triangle.



Step V: Measure the sides $\overline{PR}$, $\overline{QR}$, and angle PQR. Thus $m\overline{PR} = 4$ cm, $m\overline{QR} = 4$ cm and $m \angle PQR = 60^\circ$

Hence PQR is the required equilateral triangle.

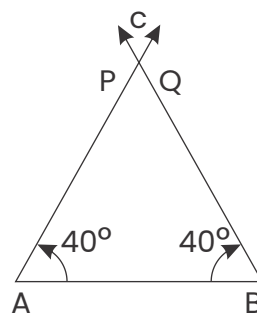
Example 2. Construct an isosceles ABC where $m\overline{AB} = 4.5$ cm
 $m \angle ABC = m \angle CAB = 40^\circ$

Given: two angles of equal measure and one side is given

Steps of construction:

Step I: Draw $\overline{AB}$ of measure 4.5 cm.

Step II: Place the central point of the protractor at point A in such a way that $\overline{AB}$ coincides with the base line of the protractor. Read it up to 40° and mark at point P.



Step III: Place the central point of the protractor at point B in such a way that $\overline{AB}$ coincides with the base line of the protractor. Read it up to 40° and mark it at point Q.

Step IV: From points A and B, draw $\overrightarrow{AP}$ and $\overrightarrow{BQ}$ to intersect each other at a point.

Step V: Mark the point as C. In this way, we get an isosceles triangle ABC. Now measure the remaining two sides $\overline{AC}$ and $\overline{BC}$ with the help of a ruler. Also, measure the third angle ACB with the help of a protractor.

Step V: Mark the point as C. In this way, we get an isosceles triangle ABC. Now measure the remaining two sides $\overline{AC}$ and $\overline{BC}$ with the help of a ruler. Also, measure the third angle ACB with the help of a protractor.

Example 1. Construct a scalene XYZ, when $m \angle XYZ = 65^\circ$ $m \angle XZY = 45^\circ$ and $m \overline{YZ} = 5.4$ cm

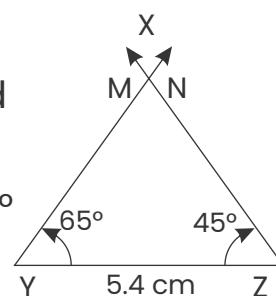
Steps of construction:

Step I: With the help of a ruler and pencil, draw $\overline{YZ}$ of 5.4 cm in length.

Step II: Place the protractor at point Y. Read it and mark a point up to 65° and mark it as M.

Step III: From point Z, read the protractor up to 45° and mark at a point N.

Step IV: Through points Y and Z, draw $\overrightarrow{YM}$ and $\overrightarrow{ZN}$ to intersect each other at point X.



Step V: Measure XY , XZ , and angle YXZ .

$m \overline{XY} = \underline{\hspace{1cm}} \text{ cm}$, $m \overline{XZ} = \underline{\hspace{1cm}} \text{ cm}$, and $m \angle YXZ = 70^\circ$

Thus, XYZ is the required scalene triangle

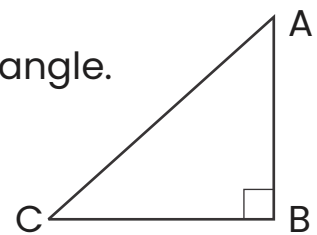
Define Hypotenuse of a right-angled triangle.

In a right triangle, the side opposite to the right angle is called hypotenuse.

Note: Hypotenuse is the longest side in a right angled triangle.

In figure, ABC is a right triangle, where B is a right angle.

$\overline{AC}$ or $\overline{CA}$ is hypotenuse, as it is opposite to right angle B .



Use protractor and ruler to construct a right-angled triangle when the measures of two angles and the measure of the side between them is given

Example:

Construct a right-angled ABC , where $m \angle BAC = 50^\circ$, $m \angle ABC = 40^\circ$ and $m \overline{AB} = 3.5 \text{ cm}$.

Steps of construction:

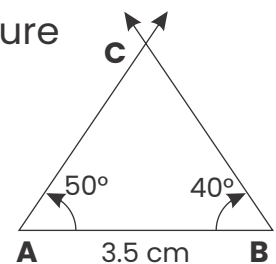
Step I: Draw $\overline{AB}$ of measure 3.5 cm .

Step II: Take point A as center, draw $\angle BAC$ of measure 50°

Step III: Take point B as center, draw $\angle ABC$ of measure 40° .

Step IV: The point where arms of both the angles meet at point C , form right angle.

Thus, ABC is the required right-angled triangle.



Use protractor, a pair of compasses and ruler to construct acute-angled, obtuse-angled and right-angled triangle when one angle and adjacent sides are given

Example 1: Construct an acute-angled ABC , where $m \angle ABC = 75^\circ$

$m \overline{AB} = 4.2 \text{ cm}$ $m \overline{BC} = 3.8 \text{ cm}$

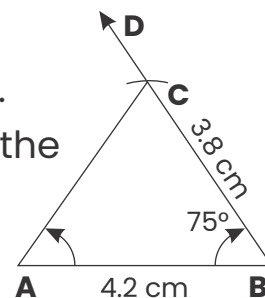
Steps of construction:

Step I: Draw $\overline{AB}$ of measuring 4.2cm.

Step II: Using protractor make $\angle ABD$ measuring 75° .

Step III: From point B, draw an arc of 3.8cm to cut the arm $\overrightarrow{BD}$ at point C.

Step IV: Draw $\overline{AC}$. We get the required acute angled $\triangle ABC$.



Example 2: Construct an obtuse angled $\triangle DEF$, when $m \angle DEF = 110^\circ$, $m \overline{DE} = 4.5$ cm and $m \overline{EF} = 4$ cm.

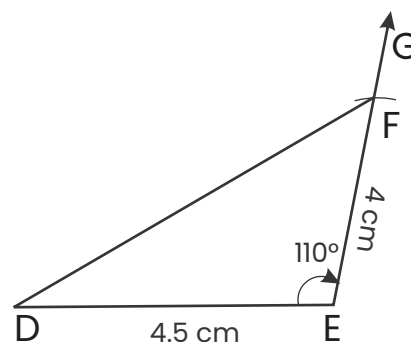
Steps of construction:

Step I: Draw $\overline{DE}$ of measuring 4.5cm.

Step II: Using protractor make $\angle DEG$ measuring 110° .

Step III: Use pair of compass, select E as centre and draw arc of 4cm radius to cut the line $\overrightarrow{EG}$ at point F. Use scale and draw $\overline{DF}$, which is the third side of the $\triangle DEF$.

Step IV: Thus, we get the required obtuse angled $\triangle DEF$.



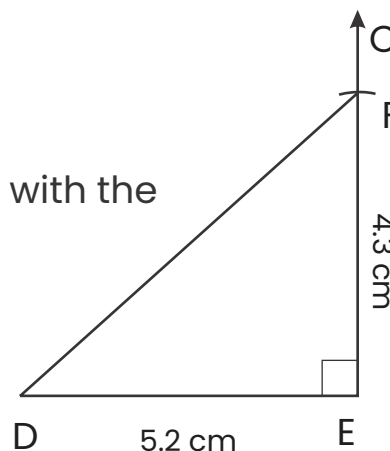
Example 3: Construct a right-angled triangle with sides $m \overline{DE} = 5.2$ cm, $m \overline{EF} = 4.3$ cm and $m \angle DEF = 90^\circ$.

Steps of construction:

Step I: Draw $\overline{DE}$ measuring 5.2cm.

Step II: From point E, draw $\overline{EC}$ a right angle with the help of protractor.

Step III: With E as centre and radius 4.3cm, draw an arc to cut $\overrightarrow{EC}$ at point F, the third vertex of $\triangle DEF$.



EXERCISE 7.5

Q1. Using a ruler and a pair of compasses, construct the following equilateral triangles.

a. $\triangle PQR$ where $m \overline{PQ} = m \overline{QR} = m \overline{RP} = 6 \text{ cm}$.

Solution:

b. $\triangle ABC$ where $m \overline{AB} = m \overline{BC} = m \overline{AC} = 5.5 \text{ cm}$.

Solution:

c. $\triangle XYZ$ where $m \overline{XY} = m \overline{YZ} = m \overline{XZ} = 6.2$ cm.

Solution:

Q2. Using a ruler and a pair of compasses, construct the following isosceles triangles.

a. $\triangle DEF$ where $m \overline{DE} = m \overline{EF} = 4$ cm; $m \overline{DF} = 8$ cm.

Solution:

b. $\triangle PQR$ where $m \overline{PQ} = 4.5 \text{ cm}$, $m \overline{QR} = m \overline{PR} = 8 \text{ cm}$.

Solution:

Q3. Using a ruler and a pair of compasses, construct the following scalene triangles.

$\triangle XYZ$ where $m \overline{XY} = 6.8 \text{ cm}$, $m \overline{YZ} = 5 \text{ cm}$ and $m \overline{XZ} = 8 \text{ cm}$.

Solution:

$\triangle ABC$ where $m \overline{AB} = 6.5 \text{ cm}$, $m \overline{BC} = 5 \text{ cm}$ and $m \overline{AC} = 2.5 \text{ cm}$.

Solution:

Q4. Using ruler, protractor and a pencil to construct the following equilateral, isosceles and scalene triangles. Also measure the remaining two sides and one angle in each triangle.

a. An equilateral $\triangle ABC$ where $m \overline{AB} = m \overline{BC} = m \overline{CA} = 7.4 \text{ m}$

Solution:

b. An isosceles $\triangle PQR$ where $m \overline{PQ} = m \overline{QR}, = 8 \text{ cm}$ and $m \angle PQR = 90^\circ$.

Solution:

c. An equilateral $\triangle XYZ$ where $m \overline{XY} = 4.8 \text{ cm}$ and $m \angle XYZ = m \angle ZXY = 60^\circ$

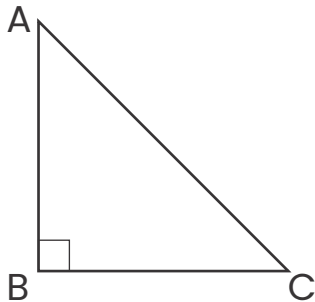
Solution:

d. An isosceles $\triangle EFG$ where $m \angle EGF = 65^\circ$, $m \overline{EG} = 6.6$ cm and $m \angle FEG = 65^\circ$.

Solution:

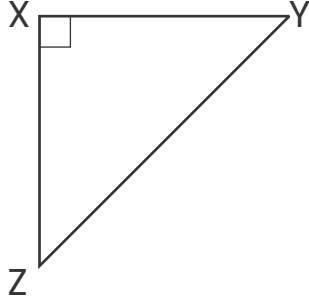
Q5. Look at the following right angled triangles. Name the hypotenuse in each triangle and measure its length.

(i)

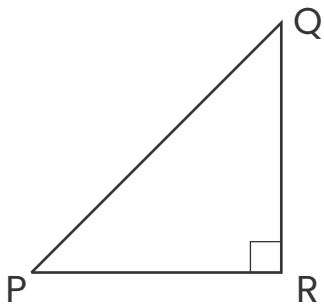


Solution:

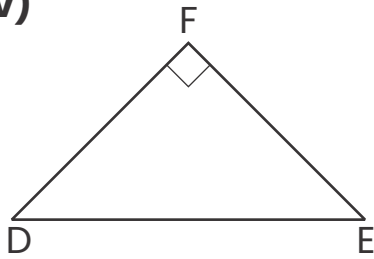
(ii)

**Solution:**

(iii)

**Solution:**

(iv)



Solution:

Q6. Using a ruler and protractor to construct the following triangles. Measure the length of remaining two sides and one angle of the triangle

$\triangle PQR$ where $m \overline{PQ} = 6\text{cm}$, $m \angle QPR = 65^\circ$, $m \angle PQR = 45^\circ$

Solution:

$\triangle ABC$ where $m \angle BAC = 85^\circ$, $m \angle ABC = 45^\circ$, $m \overline{AB} = 6.8\text{ cm}$

Solution:

$\triangle GHI$ where $m \overline{GH} = 7.3 \text{ cm}$, $m \angle GHI = 95^\circ$ $m \angle HGI = 35^\circ$

Solution:



Q7. Use protractor, a pair of compasses and ruler to construct the following triangles:

Acute-angled $\triangle PQR$, when $m \angle QPR = 85^\circ$, $m \overline{PQ} = 5.6 \text{ cm}$ and $m \overline{PR} = 6.4 \text{ cm}$.

Solution:

Right-angled $\triangle ABC$, when $m \angle ABC = 90^\circ$, $m \overline{AC} = 3 \text{ cm}$ and $m \overline{BC} = 4 \text{ cm}$.

Solution:

Obtuse-angled $\triangle GHI$, $m \angle IHG = 130^\circ$, $m \overline{HI} = m \overline{HG} = 7 \text{ cm}$.

Solution:

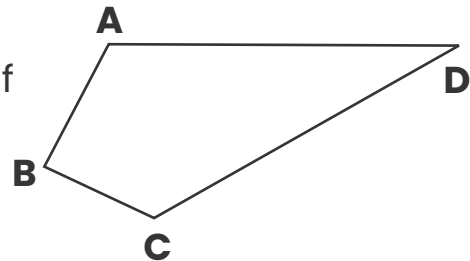
QUADRILATERALS

A closed figure with four sides is called a quadrilateral

In the figure, quadrilateral ABCD has four sides $\overline{AB}$, $\overline{BC}$, $\overline{CD}$ and $\overline{AD}$.

It has four angles A, B, C, and D.

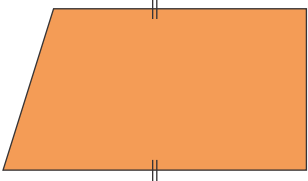
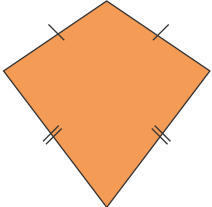
The sum of measure of all four angles of quadrilateral is equal to 360° .



Recognize the kinds of quadrilateral.

The kinds of a quadrilateral are:

Quadrilateral	Figure	Properties
Square		• All the four sides are equal. • Opposite sides are parallel. • Each angle is of 90°.
Rectangle		• Opposite sides equal • Opposite sides are parallel • Each angle is of 90°
Parallelogram		• Opposite sides equal • Opposite sides are parallel • Opposite angles are equal • None of the angles measures 90°
Rhombus		• Four equal sides • Opposite sides are parallel • Opposite angles are equal • None of the angles measures 90°

Trapezium		Only one pair of opposite parallel sides
Kite		Has two distinct pairs of adjacent sides that are equal in length.

Use protractor, set squares and ruler to construct square and rectangle with given sides.

Example 1: With the help of ruler and protractor construct a rectangle of sides 4.5 cm and 2.5 cm.

Steps of construction:

Step I: Use ruler, draw $\overline{EF}$ of 4.5 cm.

Step II: At point F, draw $\overline{EFB}$ 90° with the help of protractor.

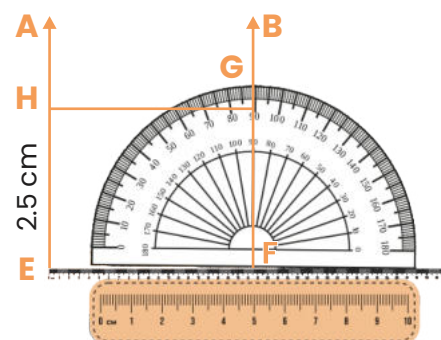
Step III: Using ruler cut $\overline{FB}$ such that $\overline{FG}$ 2.5 cm.

Step IV: At point E, draw $\overline{FEA} = 90^\circ$ with the help of protractor.

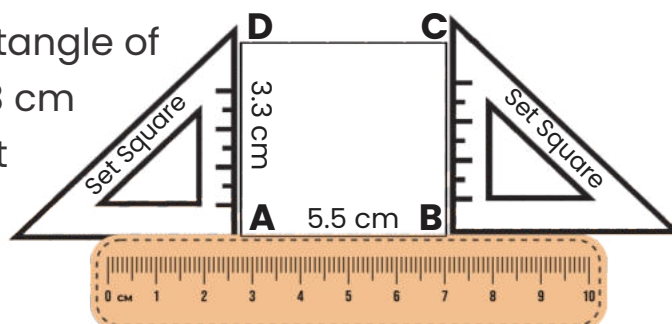
Step V: Using ruler, cut $\overline{EA}$ such that $\overline{EH}$ of 2.5 cm.

Step VI: Draw $\overline{GH}$ by joining the points G and H, i.e., draw $\overline{GH}$ of 4.5 cm.

Hence quadrilateral EFGH is the required rectangle.



Example 2: To construct a rectangle of length 5.5 cm and breadth 3.3 cm with the help of ruler and a set square.



Steps of construction:

Step I: Use ruler, draw $\overline{AB}$ of 5.5 cm length.

Step II: Place both set squares on $\overline{AB}$, such that their right-angled perpendicular sides are parallel to each other.

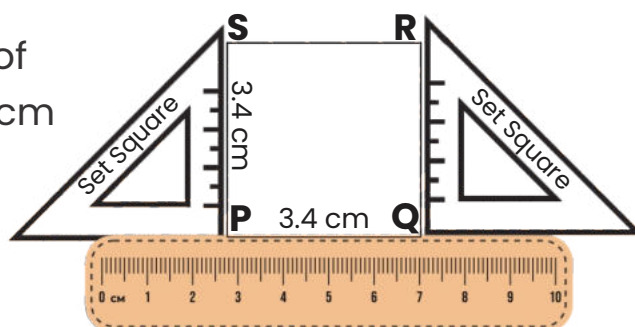
Step III: With the help of set squares, construct a perpendicular at point A. Cut it 3.3 cm and mark it point D.

Step IV: Construct perpendicular with the help of set squares at point B. Draw another perpendicular $\overline{BC}$ and cut it 3.3 cm length and mark it point C.

Step V: Join point C and D with the help of ruler and draw $\overline{CD}$.

Thus quadrilateral ABCD is the required rectangle.

Example 3: Construct a square of which the length of a side is 3.4 cm with the help of ruler and a set square.



Steps of construction:

Step I: Use ruler, draw $\overline{PQ}=3.4\text{cm}$ long.

Step II: Place both set squares on PQ, such that their right-angled perpendicular sides are parallel to each other.

Step III: With the help of set square, draw a perpendicular on point P. Measure it 3.4 cm and mark it as point S.

Step IV: Construct another perpendicular at point Q. Cut it 3.4 cm and mark it point R.

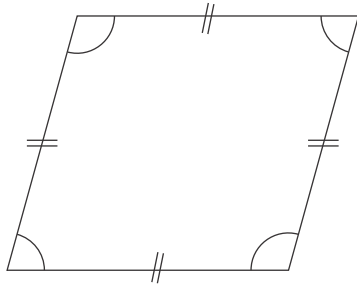
Step V: Use ruler, join point R and S then draw $\overline{RS}=3.4\text{cm}$ long.

Thus quadrilateral PQRS is the required square.

EXERCISE 7.6

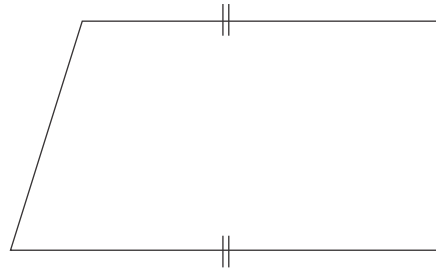
Q1. Recognize the names of the given quadrilateral.

(i)



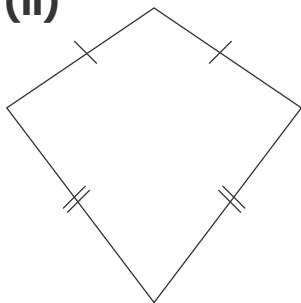
Solution:

(iii)



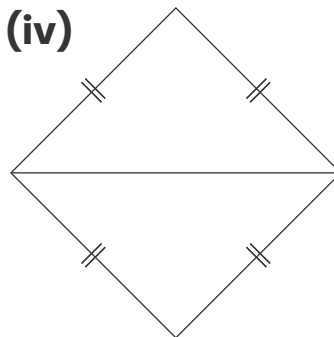
Solution:

(ii)



Solution:

(iv)



Solution:

Q2. Construct rectangle with sides given below with the help of a ruler and a protractor.

(i) 6 cm, 2 cm

Solution:

(ii) 3 cm, 5 cm

Solution:

Q3. Construct a square with a side given below with the help of a ruler and a protractor.

(i) 3.5 cm

Solution:

(ii) 4.5 cm

Solution:

Q4. Construct rectangles with sides given below with the help of a ruler and set squares.

(i) 4.5 cm, 2.8 cm

Solution:

(ii) 5 cm, 3 cm

Solution:

Q5. Construct a square with a side given below with the help of a ruler and a set squares.

(i) 6 cm

Solution:

(ii) 5.3 cm

Solution: